

MitoChime

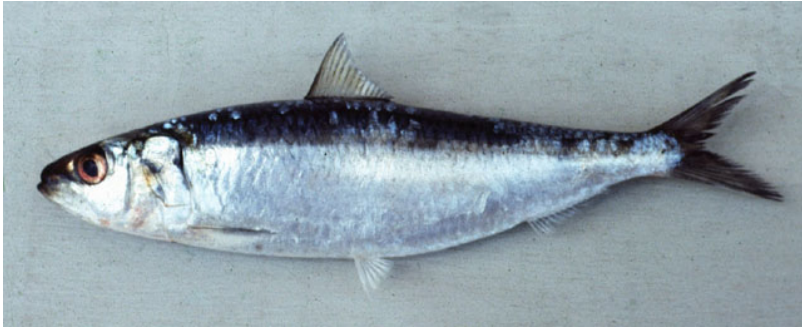
A Machine Learning Pipeline for Detecting PCR-Induced Chimeras in Mitochondrial Illumina Reads

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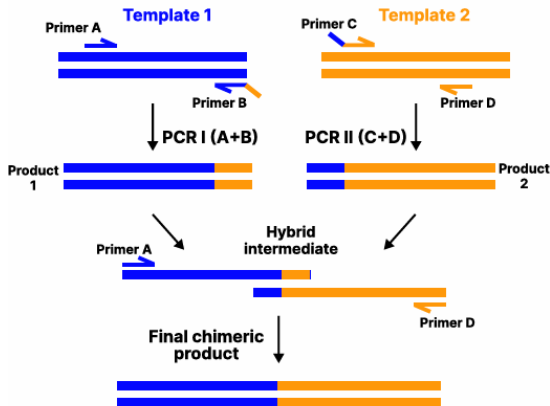
The *Sardinella lemuru*



Sardinella lemuru. Photo by T. Gloerfelt-Tarp, via FishBase

https://www.fishbase.se/FieldGuide/FieldGuideSummary.php?genusname=Sardinella&speciesname=lemuru&c_code=344

Problem: PCR-Induced Chimeras



Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.

Implication

Assembly quality

- Fragmented or failed mitochondrial assemblies
- Reduced reliability of mitochondrial genomics workflows

Research impact

- Limited downstream analyses in fisheries, population genetics, and evolution
- Wasted sequencing effort, computing time, and researcher labor

Existing approaches

Tool / Approach	Main focus	Gap for this study
UCHIME / UCHIME2	Amplicon chimeras	Reference/abundance dependent
DADA2	ASV-level filtering	Not read-level mtDNA filtering
CATCH	16S rRNA ensemble ML	Not for mtDNA; inconsistent
ChimPipe	RNA-seq fusion/chimera detection	Not PCR-induced mtDNA artifacts
MitoChime	Read-level mtDNA chimeras	Pre-assembly ML/DL filtering

Most existing tools target amplicon, ASV, or RNA-seq chimeras; MitoChime targets PCR-induced mitochondrial short-read artifacts.

Can machine learning detect **PCR-induced chimeric reads** in mitochondrial Illumina data, and can filtering these reads **improve downstream genome assembly?**

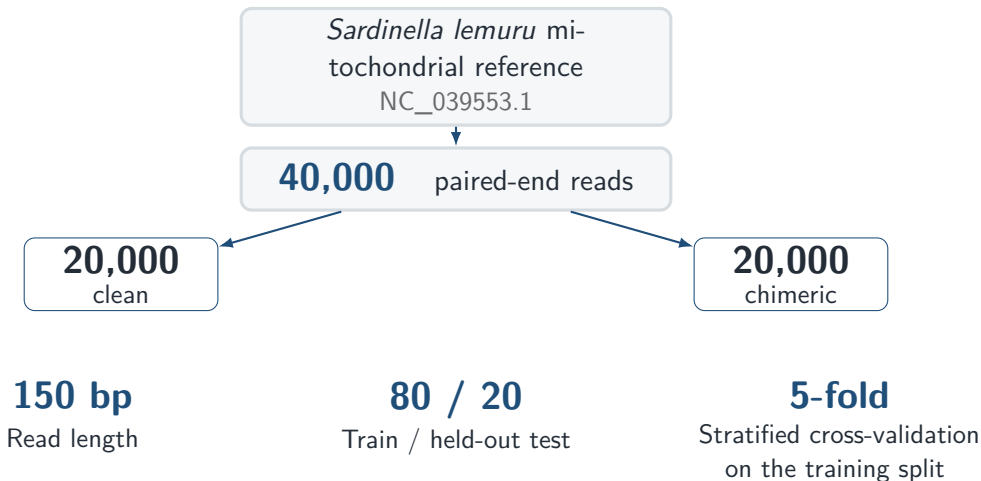
Input

Simulated clean and chimeric paired-end Illumina reads from *S. lemuru*

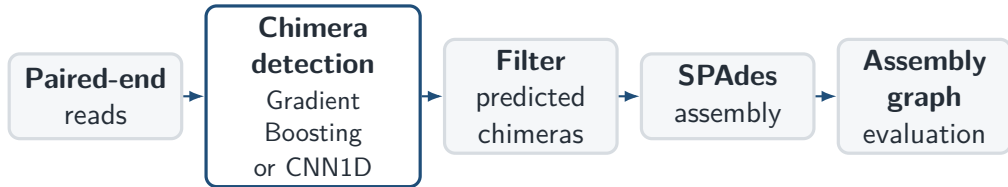
Output

A modular pipeline that predicts and filters suspected chimeric reads

Experimental Setup



MitoChime Workflow



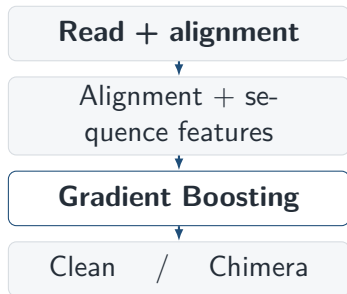
Predicted chimeric reads are removed **before assembly**.

Downstream impact is evaluated through the resulting assembly graph structure.

Two Approaches to Chimera Detection

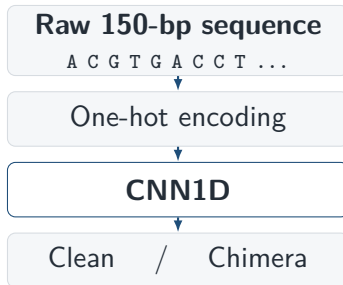
Gradient Boosting

Feature-based



CNN1D

Sequence-based



Chimera Detection Performance

Held-out test performance

Model	Precision	Recall	F1	ROC-AUC
Gradient Boosting	0.937	0.663	0.777	0.846
CNN1D	0.872	0.959	0.912	0.961

Gradient Boosting

- Very high precision: **0.937**
- Lower recall: **0.663**
- **179 false positives**
- **1,348 false negatives**

Gradient Boosting: fewer false positives

CNN1D

- Very high recall: **0.959**
- Higher F1 and ROC-AUC
- **561 false positives**
- **165 false negatives**

CNN1D: fewer false negatives

Effect on Downstream Assembly

High Coverage and Severe Chimera Contamination

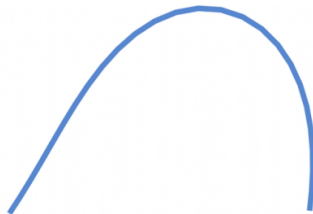
20,000 read pairs — 50% chimeras



Unfiltered



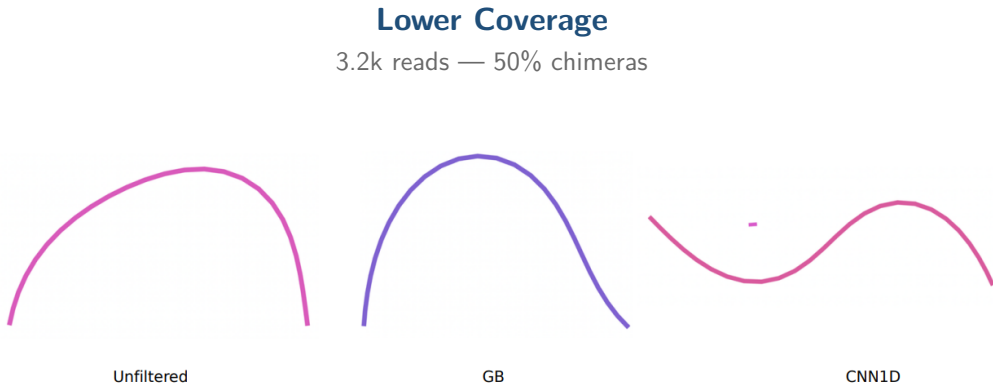
GB



CNN1D

At high read depth and severe chimera contamination, both filtering approaches dramatically reduced assembly graph complexity.

Effect on Downstream Assembly



At lower read depth, aggressive CNN1D filtering can remove too many valid reads and reduce the amount of read support available for assembly.

Overall Interpretation

Main Findings

- PCR-induced chimeric reads can be detected using both **alignment-derived features** and **raw nucleotide sequences**.
- **CNN1D achieved the strongest overall classification performance**, with an F1 score of **0.912** and ROC-AUC of **0.961**.
- Chimera filtering can substantially reduce **assembly graph complexity**, but its effectiveness depends on **available read depth**.

Choosing the Filtering Strategy

At Low Coverage

Gradient Boosting may be safer

- Higher precision
- Preserves more clean reads
- Maintains assembly support

At Moderate–High Coverage

CNN1D may be more advantageous

- Higher recall
- Removes more chimeras
- Remaining reads can compensate

Practical Implication

- MitoChime can serve as a **pre-assembly quality-control step** for mitochondrial genome assembly.
- Users can select between **Gradient Boosting** and **CNN1D**, as well as adjust the **filtering threshold**, depending on the characteristics of their dataset.

Limitations

- The study used **simulated sequencing reads**.
- Evaluation was performed using a **single mitochondrial reference** from *Sardinella lemuru*.
- Simulation cannot fully reproduce real-world sequencing conditions, including:
 - Uneven coverage
 - Real sequencing errors
 - Contaminants
 - Nuclear mitochondrial sequences
 - Library-specific artifacts
 - Differences among taxa and laboratory protocols

Future Work

Future work will include:

- Validation using **real sequencing libraries**
- Evaluation across **additional species and mitochondrial references**
- Testing under **different sequencing error profiles**
- Evaluation across **different library preparation conditions**
- **Dataset-specific threshold calibration**

Final Takeaway

- Detecting PCR-induced chimeras should not be treated only as a **classification problem**.
- The effect of filtering on **downstream assembly** and the amount of **available read support** must also be considered when choosing the appropriate filtering strategy.
- **MitoChime shows potential as a practical pre-assembly quality-control tool for mitochondrial genome assembly workflows.**